

HARISH DERANGULA

Senior Python Developer | 📞 +91 7013799924 | ✉ harish.derangula99@gmail.com

PROFESSIONAL SUMMARY:

- Senior Python Software Engineer with 7+ years of IT experience, including extensive hands-on experience in Python web application development.
- Generative AI Engineer with expertise in RAG pipelines, LLM integration (Llama 3/LangChain), and Spark-based data processing for enterprise knowledge assistants.
- Strong expertise in building scalable applications using Django, Flask, Cherrypy, and developing RESTful APIs with Django REST Framework.
- Proven experience in designing micro services architectures, deploying on AWS (Lambda, EC2, S3), and managing CI/CD pipelines using Jenkins, Docker, and Kubernetes.
- Hands-on experience in developing automation frameworks, writing unit/component test cases, and performing REST API testing using Postman.
- Skilled in handling large and unstructured datasets using Pandas, along with writing optimized SQL queries for CRUD operations across PostgreSQL, MySQL, and MongoDB.
- Strong foundation in Object-Oriented Programming (OOP), Data Structures & Algorithms, and Software Development Life Cycle (SDLC).
- Experienced in Agile/Scrum environments, participating in code reviews, debugging production issues, and delivering clean, reusable, maintainable code.
- Proficient with Git, Jira, Linux, and cloud-native development practices.
- Known for strong problem-solving abilities, effective cross-functional collaboration, and continuous learning of emerging technologies.

TECHNICAL EXPERTISE:

Programming Languages	Python
Web Development	Django, Flask, REST APIs, CherryPy
Generative AI	Lang chain, RAG Pipeline
Cloud Technologies	AWS (Lambda, EC2, S3, SQS)
DevOps / CI-CD / Version Control	Git, GitHub, Jira, Docker, Jenkins, Kubernetes
Database Management	MySQL, MongoDB (NoSQL)
Front-End Technologies	HTML, CSS
IDEs	PyCharm, Visual Studio Code
Tools & Documentation	Stoplight, Confluence
Operating Systems	Linux, Windows
Core Concepts	Data Structures & Algorithms, Object-Oriented Programming (OOP), ORM, RESTful Web Services
Development Methodologies	Agile/Scrum, SDLC
Code Quality	Debugging & Troubleshooting, Code Optimization, Clean Code
Problem Solving	Strong analytical skills for resolving complex technical challenges
Communication	Effective collaboration with cross-functional teams and clients
Professional Strengths	Continuous Learning, Building scalable and maintainable solutions

EDUCATION:

M.Tech in Computer Science, Jawaharlal Nehru Technological University, Anantapur (2015–2017)

PROFESSIONAL EXPERIENCE:

- Module delivery Lead | Mphasis, Bangalore. August 2025 to September 2025.
- Software Developer - Python | Buzz works business service Pvt.Ltd, Bangalore. March 2025 to May 2025.
- Senior Development Engineer | ITS Cybertech India Pvt.Ltd, Pune. February 2024 to January 2025.
- Test Analyst | Infosys Ltd, Hyderabad. January 2023 to February 2024
- Associate | AXS Online Pvt. Ltd, Bangalore March 2018 to December 2022.

PROFESSIONAL SUMMARY:

Project#5:

Name: AI-Based Medical Document Validation System

Client: Confidential Healthcare / Insurance Client

Duration: March 2025 to May 2025.

Technologies- Python, Django/FastAPI, PostgreSQL, Redis, AWS (S3, EC2), Docker, Kubernetes, Linux

Key Responsibilities:

- Designed and developed RESTful APIs for secure medical document upload and validation.
- Implemented document processing pipeline for handling multi-page PDFs and scanned files.
- Integrated OCR engine to extract text data from medical documents for validation checks.
- Developed blank page detection module using image variance analysis with OpenCV.
- Implemented page number validation logic using text extraction and regex-based sequence verification.
- Built signature presence detection mechanism using image processing techniques.
- Designed asynchronous processing using Celery and Redis to handle large document volumes efficiently.
- Integrated AWS S3 for secure storage and retrieval of uploaded documents.
- Designed database schema for storing document metadata, validation results, and audit logs.
- Implemented error handling, logging, and retry mechanisms for failed processing jobs.
- Containerized application services using Docker and deployed in Kubernetes environment.
- Collaborated with QA and business teams to ensure compliance with healthcare documentation standards.

Project#4:

Name: Asset Processing Services

Duration: Feb 2024 to Dec 2024.

Technologies- Python, Docker, Django, CICD Pipelines, AWS (lambda, ec2), PostgreSQL, Micro services.

Description:

Designed and implemented an automation-driven extension service using micro services architecture to generate new assets from existing resources, enabling end-to-end request lifecycle management, real-time tracking, and service performance monitoring.

Key Responsibilities:

- Designed and developed RESTful APIs using Python and Django, delivering scalable and maintainable backend services.
- Performed manual REST API and web services testing using Postman to validate functionality and integration.
- Created unit and component test cases using Python's unit test framework to ensure code reliability and coverage.
- Identified, debugged, and resolved defects in existing codebases to improve application stability and performance.
- Participated in peer code reviews, enforcing coding standards and driving continuous quality improvements.
- Worked within an Azure-based agile development environment, actively contributing to daily Scrum ceremonies.
- Monitored and troubleshoot CI/CD pipelines, resolving build and deployment failures to maintain delivery timelines.
- Conducted security assessments using tools such as Black Duck, Checkmarks, and SonarQube to identify and remediate vulnerabilities.
- Authored and maintained comprehensive REST API documentation, supporting knowledge transfer and long-term application maintenance.

Project#3:

Name: Road side Assistance (RSA)

Duration: Feb 2023 to Jan 2024.

Technologies: Python, Django, DRF, PostgreSQL, RestAPI's, Pandas, Numpy.

Description:

Camping World is a web-based application that supports multiple services such as rentals, spare parts e-commerce, and tire rescue. The platform allows service agents to log customer complaints, create service requests, and place orders on behalf of clients, providing a centralized system for managing customer support and service operations.

Key Responsibilities:

- Developed application frameworks based on business requirements and created unit and component test cases using Python's built-in unit test module to ensure code quality.
- Created and maintained API documentation using Stoplight, updating specifications during bug fixes and feature enhancements.
- Monitored Jenkins CI/CD pipelines and validated deployments across Windows, Linux, and Docker container environments.
- Implemented multithreading in Python to improve application performance and processing efficiency.
- Performed continuous security monitoring using SonarQube, Twist lock, and Black Duck, identifying and addressing vulnerabilities.
- Assisted in analysing and supporting application deployments in Kubernetes environments.
- Debugged and resolved software defects, ensuring system stability and improved performance.
- Collaborated with cross-functional teams to design and deliver new features and innovative solutions.
- Worked closely with development teams to implement enhancements aligned with project objectives. Participated in Python codebase migration efforts to optimize application performance and reduce application size. Demonstrated working knowledge of Go concepts, including Struts, Interfaces, Go routines, Concurrency, Channels, and Error Handling.

Project#2:

Name: Aigtravel guard insurance.

Duration: June 2021 to Dec 2022.

Technologies: Python, Flask, PostgreSQL, Bit bucket, Rest Api, Agile, Scrum, Unit test.

Description:

AIG Travel Guard is a web-based platform that allows users to explore, purchase, and manage travel insurance policies while providing assistance services to protect travellers from unexpected disruptions during their journeys.

Key Responsibilities:

- designed and developed backend services using Python and Flask to support travel insurance policy management and assistance workflows.
- Built and integrated RESTful APIs for policy purchase, customer on boarding, and claims-related operations.
- Implemented business logic to handle user journeys including policy exploration, purchase, and management.
- Developed and executed unit test cases to validate API endpoints and application functionality, ensuring high code quality.
- Integrated database operations using MySQL to manage customer data, policy details, and transaction records.
- Monitored application performance and resolved production issues using logging and debugging tools.
- Developed and maintained server-side application logic using Python and Flask, ensuring scalable and reliable backend services.
- Wrote unit tests to validate Flask endpoints and core application logic, improving code quality and reliability.

Project#1:

Name: Trade Web

Duration: April 2018 to May 2021

Technologies: Python, Flask, MySQL, Restapi.

Description:

Trade Web is a cloud-based SaaS platform that helps banks manage global trade operations by providing tools for trade finance, payables, receivables, and cash management. It enables banks to offer these services to their customers anytime, anywhere through a single integrated platform.

Key Responsibilities:

- Collaborated with stakeholders to understand business requirements and translate them into technical features and application functionality.
- Developed and maintained server-side application logic using Python and Flask, ensuring scalable and reliable backend services.
- Wrote unit tests to validate Flask endpoints and core application logic, improving code quality and reliability.
- Monitored application performance and resolved production issues using logging, monitoring, and alerting tools.